

FINAL CAPSTONE PROJECT

Guidelines & Evaluation Rubric

Machine Learning & Data analysis Track

To ensure that our training programs provide practical, industry-oriented experience, every course will conclude with a Capstone Project that reflects the skills and knowledge acquired throughout the program.

1

MANDATORY REQUIREMENTS

- The project must solve a real-world problem or address a meaningful use case whenever possible.
- Projects may be completed in groups of up to **four** members.
- Students should apply the concepts learned during the course, including Data Analysis, Feature Engineering, and Classical and/or Advanced Machine Learning techniques (e.g., regression, classification, clustering, ensemble methods, or deep learning).
- The final solution must include a local deployment using either Streamlit or Flask.
- The complete project must be uploaded to GitHub with clean code organization and comprehensive documentation.
- Each student must publish their project on LinkedIn to showcase their work professionally and support future career opportunities.

🚩 CERTIFICATE REQUIREMENT

Completing, documenting, and publishing the project is a mandatory requirement for receiving the course completion certificate.

2

REQUIRED DELIVERABLES

- Public GitHub repository containing the complete source code.
- README file with project description, setup instructions, dependencies, and usage guidelines.
- Presentation slides for the final discussion session.
- A working Streamlit or Flask application.
- LinkedIn post
- GitHub repository link, submitted through the dedicated Classroom assignment created for this purpose.
- Repository submission deadline: **12:00 AM (midnight) on the day immediately before the discussion/presentation day** — not the morning of the discussion itself.

🚩 GITHUB SUBMISSION DEADLINE

Each team must submit its GitHub repository link by midnight (12:00 AM) on the night before the discussion/presentation day, using the Classroom assignment created for this submission. Make sure the repository is accessible and contains all relevant work before submitting.

3 PROJECT EXPECTATIONS

- Clearly define the problem statement and its importance.
- Perform data collection, cleaning, preprocessing, and exploratory data analysis when applicable.
- Develop, train, and evaluate suitable Machine Learning models.
- Justify model choices and evaluation metrics.
- Build a simple, user-friendly local deployment interface.
- Write clean, modular, and well-documented code.
- Discuss limitations and potential future improvements.

4 ATTENDANCE & WORKDAY COMMITMENT

- Everyone is expected to arrive on time at the beginning of the workday.
- Everyone is expected to stay until the very end of the workday, which is **9:00 PM**.
- Team members should not leave early unless there is an approved, valid reason.

🚩 ATTENDANCE REQUIREMENT

Arriving on time and remaining until 9:00 PM every workday is mandatory for all team members, except where an absence or early departure has been approved in advance.

5 FINAL PRESENTATION & DISCUSSION DAY

The final discussion session will begin at **4:00pm on Wednesday [14-10-2026]**. Each team must deliver a presentation and perform a live demonstration of their project.

- Problem statement and motivation.
- Dataset description and preprocessing pipeline.
- Methodology and technical approach.
- Machine Learning techniques used (e.g., feature engineering, model selection, hyperparameter tuning, ensembling).
- Model evaluation metrics and results.
- System architecture and deployment process.
- Challenges encountered and solutions implemented.
- Future work and possible improvements.
- Live demonstration of the final application.

Group of 4

50 min

Presentation Guidelines

- **Timekeeper:** Every group must assign one team member as timekeeper, responsible for monitoring the presentation time and making sure the group stays within the allocated time.
- **Everyone participates:** All team members must be involved and present/ready to participate when the team is presenting. The presentation should clearly reflect that the work was done collaboratively.
- **Team introduction:** At the beginning of the presentation, the team should briefly introduce themselves and identify the members of the team.
- **Introduce the presentation:** The team should briefly introduce the topic/project and explain what they will cover before starting the main content.
- **Clear voice:** Each presenter must speak loudly and clearly enough to be heard by students sitting in the last row of the classroom.
- **Clear slides:** Slides should be readable and understandable from the back of the classroom. Avoid overly small fonts, excessive text, or cluttered slides.
- **Professional delivery:** Presenters should maintain good eye contact, avoid simply reading the slides word-for-word, and communicate confidently and clearly.
- **Smooth transitions:** Team members should know when they are expected to speak and transition smoothly from one presenter to another.

6 EVALUATION RUBRIC · 100 POINTS

Criterion	Description	Points
Problem Definition	Clarity, relevance, and real-world value of the problem.	10
Data Analysis & Preprocessing	Quality of EDA, preprocessing, and data understanding.	15
Model Development	Appropriate use and evaluation of Machine Learning methods.	20
Deployment	Functionality and usability of the Streamlit or Flask application.	15
Code Quality & Documentation	Organization, readability, modularity, and GitHub documentation.	15
Presentation & Communication	Quality of slides, explanations, and responses to questions.	15
Live Demo	Successful demonstration of the final product.	10

7 FINAL NOTE

Students are encouraged to be creative and integrate multiple concepts covered during the course. The Capstone Project should serve as a portfolio-quality project that demonstrates both technical and communication skills.